What Can Binary Response Patterns Reveal About Benchmark Contamination? An IRT Person-Fit Characterization

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Abstract

Benchmark contamination in Large Language Models (LLMs) undermines evaluation validity, yet most detection methods require privileged access to training data, model internals, or output probabilities. We present the first systematic characterization of IRT person-fit statistics for detecting SFT-stage contamination: the framework requires only binary response patterns from the target model—no access to weights, gradients, logits, or training data—combined with externally calibrated 4PL item parameters from a large-scale item bank; diagnostic θ-anchoring additionally requires a clean baseline. A contaminated LLM produces aberrant response patterns analogous to a test-taker with item preknowledge. Using ℓ_z^* in a difference-in-differences design ($\Delta\ell_z^*$) across three 7–8B model families (n=10 independent seeds per condition; n=6 for Mistral) on MMLU (14,042 items) and ARC-Challenge (295 items), we identify three behavior boundaries: (1) model-specific θ-shift dynamics linking baseline ability to detection sensitivity, where higher-ability models show weaker free-θ detection due to θ-shift absorption; (2) a detection-diagnosis tradeoff where simpler statistics yield stronger detection but lack ℓ_z^*'s diagnostic capabilities (seen/unseen decomposition, mechanistic θ-shift analysis); and (3) an ability-dependent non-monotonic dose-response, most pronounced in high-ability models (Qwen: $|d|$ peaks at 25%, 4.3$\times$ decline to 50%; Llama: directionally consistent, attenuated), empirically confirming θ-shift absorption. A preliminary single-model observation suggests ℓ_z^* patterns may persist after GRPO alignment, though this requires further investigation. These findings establish when and why binary response patterns carry contamination information beyond aggregate accuracy.

1 Introduction

Benchmark contamination, the unintended inclusion of evaluation data in training corpora, poses a growing threat to the integrity of large language model (LLM) evaluation (Dodge et al., 2021; Magar and Schwartz, 2022; Sainz et al., 2023). As training corpora and public benchmarks both expand, the probability of overlap increases structurally (Jacovi et al., 2023; Chang et al., 2024). When contamination occurs, benchmark scores no longer reflect genuine capability, undermining the primary mechanism by which the community compares and selects models.

A range of detection methods have been proposed, but all require some form of privileged access beyond the binary response record. N-gram overlap methods need the training corpus (Dodge et al., 2021; Magar and Schwartz, 2022). Membership inference approaches require token-level probabilities (Shi et al., 2024; Deng et al., 2024). Probing methods rely on crafted prompts (Golchin and Surdeanu, 2024a; Oren et al., 2024). Performance-based methods require multiple test variants or output logits (Dekoninck et al., 2024; Zawalski et al., 2026). These methods form a detection frontier ordered by information access. No existing method operates using only binary response patterns, without model weights, gradients, logits, or training data—though, as we show, effective detection additionally requires externally calibrated item parameters.

In this work, we focus on SFT-stage contamination—where benchmark answers are explicitly included in fine-tuning data—for three reasons. First, controlled experiments require known ground-truth contamination sets, achievable through fine-tuning with specific benchmark items. Second, SFT-stage contamination is increasingly relevant as fine-tuning-as-a-service platforms proliferate, creating new pathways for benchmark data to enter training mixtures. Third, SFT provides a methodologically clean setting to characterize detection boundaries before extending to noisier pretraining scenarios.

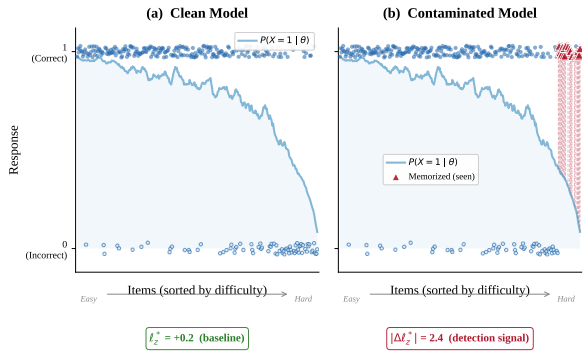


Figure 1: A clean model's responses track the IRT-predicted probability curve (left); a contaminated model shows anomalous successes on hard items (right, red diamonds), producing a large $|\Delta\ell_z^*|$ signal.

The need for detection from binary response patterns grows more pressing as models evolve. Wang et al. (2025) show that all ten log-probability-based methods they evaluate perform at chance level on models trained with reinforcement learning (RL), because RL reshapes internal token distributions while preserving binary correctness patterns. This motivates a fundamental question: *what can binary response patterns reveal about contamination, and under what conditions?*

We observe that a contaminated LLM is functionally analogous to a test-taker with item preknowledge: both produce response patterns in which difficult items are answered correctly through memorization rather than genuine ability. This connection points to person-fit statistics, developed over four decades for cheating detection in educational testing (Drasgow et al., 1985; Meijer and Sijtsma, 2001; Karabatsos, 2003; Sinharay, 2017). The standardized statistic ℓ_z^* (Snijders, 2001) measures how well a response pattern conforms to Item Response Theory (IRT) model predictions. Figure 1 illustrates this analogy. Despite the natural correspondence, no prior work has applied person-fit statistics to LLM benchmark contamination.

Rather than proposing ℓ_z^* as a ready-to-deploy detection tool, we systematically characterize the behavior boundaries of IRT person-fit statistics for LLM contamination detection. Our framework operates at three layers: (i) pre-calibrated 4PL item parameters from the PSN-IRT item bank (Zhou et al., 2026), an external resource derived from 13,000+ models; (ii) binary response patterns obtained from the target model without access to weights or logits; and (iii) retrospective statistical inference requiring no model modification. Using controlled contamination experiments across three 7–8B model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B; n=10 seeds each, n=6 for Mistral) with pre-calibrated 4PL item parameters from PSN-IRT (Zhou et al., 2026), we identify three IRT-specific findings on MMLU (Hendrycks et al., 2021) and ARC-Challenge:

1. **Model-specific θ-shift dynamics.** Baseline ability modulates detection sensitivity through θ-shift absorption: lower-ability models (Mistral, 52.4%) show stronger signals ($|d|$=3.50) than higher-ability models (Qwen, 67.8%, $|d|$=1.90) at 50% contamination. ARC-C reverses this pattern through a ceiling effect. Fixed-θ estimation restores monotonicity for models with stable baselines.
2. **Detection-diagnosis tradeoff.** Simpler statistics yield stronger detection: accuracy z-scores produce $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics, and HT generally exceeds ℓ_z^*. Yet the diagnostic power reverses: only ℓ_z^* supports seen/unseen decomposition, mechanistic θ-shift analysis, and sensitivity to response-pattern structure (57/60 vs. 4/60 flagged cells on real-world data).
3. **Ability-dependent non-monotonic dose-response.** Under free-θ estimation, Qwen shows a pronounced inverted-U ($|d|$: 6.58 $\rightarrow$ 8.22 $\rightarrow$ 1.90); Llama shows an attenuated but directionally consistent pattern, with amplitude scaling with θ-shift sensitivity.

A preliminary N=1 observation suggests ℓ_z^* patterns may persist after GRPO alignment (§5).

Our contributions are threefold: (1) We provide the first systematic characterization of IRT person-fit behavior for LLM contamination, identifying boundary conditions (θ-shift, benchmark size, model ability) that determine when and why detection succeeds or fails. (2) We establish that $\Delta\ell_z^*$ reliably detects SFT-stage contamination across three model families (all bootstrap 95% CIs exclude zero, n=10). (3) We document the detection-diagnosis tradeoff, clarifying what binary response patterns can and cannot reveal.

2 Related Work

Benchmark contamination detection. Contamination detection methods span a wide range of information requirements. Corpus-search methods match benchmark items directly against pretrain-

ing data (Dodge et al., 2021; Magar and Schwartz, 2022), but require full corpus access unavailable for closed models. Membership inference attacks (MIA) use token-level log-probabilities to distinguish training from unseen examples (Shi et al., 2024; Deng et al., 2024). Generation-based probes elicit benchmark content via prompting (Golchin and Surdeanu, 2024a,b; Oren et al., 2024). Performance-based methods compare accuracy on target and reference sets (Dekoninck et al., 2024; Zhang et al., 2024; Zawalski et al., 2026), requiring multiple evaluations or in-context learning passes. Zhou et al. (2023) and Sainz et al. (2023) argue for systematic contamination measurement, Jacovi et al. (2023) propose mitigation strategies, and Bordt et al. (2025) show that contamination can be "forgotten" under sufficient continued training. Han et al. (2026) quantify contamination in psychometric evaluations of LLMs, demonstrating that memorization effects extend beyond standard benchmarks. Kattamuri et al. (2025) take a mechanistic interpretability approach, using activation patterns to distinguish recall-based from reasoning-based responses; our work instead operates purely on behavioral response patterns without access to model internals.

A critical limitation unites all token-level methods: Wang et al. (2025) show that RL post-training renders MIA-based detectors no better than random across 10 methods and 4 benchmarks, because RL reshapes token distributions while largely preserving binary correctness patterns. Tao et al. (2026) and Zhang et al. (2026) propose RL-specific solutions but require generation-level access. Our approach requires no access to model weights, logits, or training data—only binary response patterns and externally calibrated item parameters from PSN-IRT (Zhou et al., 2026).

IRT for LLM evaluation. Item Response Theory has seen growing adoption in NLP and ML evaluation. Martínez-Plumed et al. (2019) applied IRT to analyze machine learning classifiers at the instance level, establishing that item difficulty and discrimination parameters capture meaningful variation across models. Lalor et al. (2019) proposed estimating IRT parameters from DNN-generated response patterns, removing the need for human annotators. Rodriguez et al. (2021) demonstrated that IRT-informed item weighting improves leaderboard rankings over raw accuracy. Polo et al. (2024) used IRT-based item selection to reduce benchmark sizes by 100× while preserving model rankings, and Li et al. (2025) developed adaptive testing procedures for efficient LLM evaluation. Most directly relevant, Zhou et al. (2026) calibrated 4PL item parameters for over 13,000 models across 11 benchmarks, providing the item bank our framework builds on. All of these works use IRT for ability estimation or benchmark efficiency; none apply *person-fit* statistics for contamination detection.

Person-fit statistics in psychometrics. Person-fit analysis has a long history in educational measurement for detecting aberrant test-taking behavior (see Embretson and Reise, 2000, for a textbook treatment). Levine and Drasgow (1988) formalized optimal appropriateness measurement; Drasgow et al. (1985) introduced ℓ_z, a standardized log-likelihood residual; Snijders (2001) derived the corrected ℓ_z^* that accounts for ability estimation uncertainty. Karabatsos (2003) compared 36 person-fit statistics and found ℓ_z among the most powerful for detecting cheating; we select ℓ_z^* for its theoretical grounding and HT (Van der Flier, 1982) as a nonparametric complement, following their top-ranked performance in that comparison. Reise (1990) and Meijer and Sijtsma (2001) provide systematic evaluations of person-fit methodology. Sinharay (2017) showed that ℓ_z^* is particularly effective when the set of compromised items is known, a setting analogous to benchmark contamination where memorized items constitute the compromised set. Our work transfers this psychometric toolkit to LLM evaluation, treating each model as an examinee whose item-level response pattern may reveal memorization.

3 Method

3.1 IRT Person-Fit Framework

We adopt the four-parameter logistic (4PL) IRT model to characterize contamination-induced aberrance in LLM response patterns. The 4PL model defines the probability that a model with ability θ answers item i correctly:

$$P_i(\theta) = c_i + (d_i - c_i)\,\frac{1}{1 + e^{-a_i(\theta - b_i)}} \qquad (1)$$

where a_i is discrimination, b_i difficulty, c_i the lower asymptote (guessing), and d_i the upper asymptote (slip). Item parameters are pre-calibrated by PSN-IRT (Zhou et al., 2026) from a large-scale leaderboard dataset; we use these external parameters

rather than re-estimating from potentially contaminated data, since our MML-EM refit shows discrimination recovery degrades to r_a=0.127 on contaminated leaderboard data.

Given item parameters and a binary response vector $\mathbf{x} = (x_1, \ldots, x_n)$, we compute the person-fit statistic ℓ_z^* (Snijders, 2001), a standardized log-likelihood residual quantifying deviation between observed and expected response patterns. The Snijders correction accounts for estimation uncertainty in $\hat{\theta}_{\text{MLE}}$, so that $\ell_z^* \sim \mathcal{N}(0,1)$ under the null (Drasgow et al., 1985; Snijders, 2001); Magis et al. (2012) provide a detailed treatment of how response model selection and ability estimation affect ℓ_z^*. Large negative ℓ_z^* indicates aberrant responding (unexpected failures on easy items or unexpected successes on hard items); large positive values indicate over-consistent patterns.

3.2 $\Delta\ell_z^*$ Difference-in-Differences Design

LLM response patterns systematically violate IRT assumptions of local independence and unidimensionality. In our empirical data, 57 of 60 model-benchmark cells yield $|\ell_z^*| > 1.96$ even absent contamination, rendering absolute thresholds uninformative. We address this through a difference-in-differences (DiD) design.

For controlled experiments with clean baselines, we compute:

$$\Delta\ell_z^* = \ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^* \tag{2}$$

where $\bar{\ell}_{z,\text{clean}}^*$ averages over n=10 clean seeds. The shared model misfit cancels, isolating contamination-specific aberrance. We report Cohen's d with percentile bootstrap 95% confidence intervals (10,000 resamples) as the primary effect-size measure; BCa correction (Appendix G) confirms that percentile CIs are conservative estimates.

For deployment without clean baselines, cross-benchmark comparison provides the reference: the target benchmark's ℓ_z^* is compared against benchmarks presumed clean from the same model. Significance thresholds derive from a nonparametric bootstrap null rather than the asymptotic $\mathcal{N}(0,1)$ reference, accommodating residual model misspecification. Simulation analysis (Appendix F.1) confirms that the differential design is robust to local dependence at realistic levels ($d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$) and maintains $\geq$98.5% power even under contamination-specific asymmetric dependence.

3.3 Experimental Setup

Models and contamination. We validate on three 7–8B model families: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3. Each is fine-tuned with QLoRA (rank = 16, $\alpha = 32$) under three conditions: clean, 15% contamination (benchmark items constitute 15% of training data), and 50% contamination. Following initial observation of non-monotonic dose-response for Qwen, we added a 25% contamination condition. We subsequently extended this to Llama to test whether the pattern generalizes; Mistral was not tested at 25% due to computational constraints. We train n=10 clean seeds for Qwen and Llama and n=6 for Mistral (four checkpoints were lost when a training server was decommissioned during the study) to establish baseline variability and construct bootstrap confidence intervals. Reported CIs reflect variability across n clean baseline seeds, as each contaminated model is a single QLoRA run.

Benchmarks. We evaluate on MMLU (Hendrycks et al., 2021) (14,042 items) and ARC-Challenge (Clark et al., 2018) (295 items), spanning two orders of magnitude in benchmark size. All 4PL item parameters are from Zhou et al. (2026). ARC-Challenge experiments were conducted on Qwen only, as the primary evaluation of cross-benchmark generalizability; extending to additional families was constrained by computational budget.

Post-RL extension. To probe behavior boundaries beyond SFT, we apply Group Relative Policy Optimization (GRPO; Shao et al., 2024) to one contaminated Qwen model (15% MMLU) for 1,500 steps, evaluating at three checkpoints. This extension is N=1 (single model family, single contamination level).

Contamination model. QLoRA SFT on benchmark items serves as a controlled proxy for SFT-stage contamination, where benchmark data is explicitly included in fine-tuning mixtures—analogous to a test-taker with item preknowledge. Our findings are scoped to SFT-based contamination and may not generalize to pretraining contamination, which operates through different learning dynamics.

Table 1: Cohen's d for $\Delta\ell_z^*$ across model families (n=10 clean seeds, bootstrap 95% CI, 10,000 resamples). All CIs exclude 0.

Model	Bench.	d (15%) [CI]	d (25%) [CI]	d (50%) [CI]
Qwen	MMLU	-6.58 [-15.2, -4.5]	-8.22 [-18.9, -5.7]	-1.90 [-5.0, -1.0]
Llama	MMLU	-2.27 [-4.6, -1.6]	-2.97 [-6.1, -2.0]	-2.50 [-5.0, -1.8]
Mistral†	MMLU	-2.64 [-7.4, -1.5]	—	-3.50 [-9.6, -2.1]
Qwen	ARC-C	$+3.16$ [2.0, 9.5]	—	$+5.30$ [3.6, 15.2]

†Mistral: n=6 seeds (wider CIs reflect reduced sample size); Qwen and Llama: n=10.

4 Results

4.1 Detection Signal Across Model Families

Table 1 presents Cohen's d effect sizes for ℓ_z^* across three model families and two benchmarks. All confidence intervals exclude zero, confirming that $\Delta\ell_z^*$ reliably separates contaminated from clean models at both contamination levels.

On MMLU, effect sizes range from $|d|$=1.90 (Qwen at 50%) to $|d|$=8.22 (Qwen at 25%), with Llama ($|d|$=2.27/2.97/2.50) and Mistral ($|d|$=2.64/3.50) falling between. On ARC-C (295 items), Qwen yields $|d|$=3.16 at 15% and $|d|$=5.30 at 50%, demonstrating that detection remains feasible even on small benchmarks when contamination is sufficiently concentrated.

Clean-model ℓ_z^* baselines diverge substantially across families: Qwen $+8.68$ (SD=0.74), Mistral $+4.02$ (SD=2.05), Llama $+0.29$ (SD=1.45), and ARC-C Qwen -5.42 (SD=0.16). Absolute ℓ_z^* thresholds would produce contradictory results: Qwen's contaminated ℓ_z^* at 15% ($+3.85$) is still positive. The $\Delta\ell_z^*$ design normalizes these divergent baselines: Cohen's d yields comparable effect sizes ($|d|$=1.90–8.22) despite order-of-magnitude baseline differences. Figure 2 provides a visual summary.

4.2 Model-Specific θ-Shift Dynamics

Contamination raises overall accuracy, which inflates $\hat{\theta}_{\text{MLE}}$; a higher $\hat{\theta}$ "explains away" anomalous successes on hard items, absorbing the contamination signal into the ability estimate. We term this *θ-shift absorption*. This mechanism interacts with baseline model ability to produce an observed trend: higher-ability models show weaker detection signals under free-θ estimation at high contamination. Figure 3 plots baseline accuracy against Cohen's $|d|$ for all model-benchmark combinations.

On MMLU, a negative correlation emerges between baseline accuracy and $|d|$ at 50% contamination: Mistral (52.4% accuracy, $|d|$=3.50) > Llama (59.3%, $|d|$=2.50) > Qwen (67.8%, $|d|$=1.90). The explanation is geometric: a model already near the asymptotic ceiling of most item characteristic curves gains less $\hat{\theta}$ from contamination-induced accuracy boosts, but the reduced θ-shift headroom leaves less room for the ability estimate to absorb the aberrance signal. At 15% contamination, the pattern is less pronounced because the θ-shift is smaller in absolute terms, leaving a larger proportion of the signal as residual aberrance. Llama's minimal θ-shift ($\Delta\theta$=0.09) compared to Qwen's substantial shift ($\Delta\theta$=0.47) predicts attenuated non-monotonicity at higher contamination levels, which §4.4 confirms.

ARC-C reverses this pattern. Despite Qwen's high baseline accuracy on ARC-C (90.0%), contamination produces larger, not smaller, effect sizes ($|d|$=3.16 at 15%, $|d|$=5.30 at 50%). This reversal reflects a ceiling effect: with only 295 items and 90.0% baseline accuracy, the already-correct items offer minimal room for further θ inflation, so contamination manifests primarily as anomalous successes on the few remaining hard items. The small item count amplifies this concentration, producing a disproportionately large ℓ_z^* shift per contaminated item.

The baseline ℓ_z^* for ARC-C is negative (-5.42), reflecting the 4PL model's expectation of near-perfect performance; contamination pushes ℓ_z^* toward zero, producing the positive Cohen's d values in Table 1.

These findings indicate that detection sensitivity is not a fixed property of ℓ_z^* but depends on the interaction between model ability and benchmark characteristics, a consideration absent from standard person-fit applications.

Fixed-θ estimation. Table 2 compares free-θ and fixed-θ Cohen's d across all three families. The benefit of fixed-θ is strongly modulated by baseline ability. Qwen (67.8% accuracy) shows the largest amplification, because the high baseline provides substantial θ-shift headroom that free estimation exploits to absorb the signal. Mistral (52.4%) shows moderate amplification (1.6× at 50%), consistent with intermediate baseline ability. Llama (59.3%) shows no benefit from fixed-θ: its high cross-seed ℓ_z^* variance (SD=1.45 vs. Qwen's 0.16 on ARC-C) inflates the denominator, and its free-θ effect sizes vary minimally across contamination levels ($|d|$=2.27 → 2.97 → 2.50). The Llama result is itself informative: fixed-θ estimation is indicated for models with stable clean baselines and high θ-

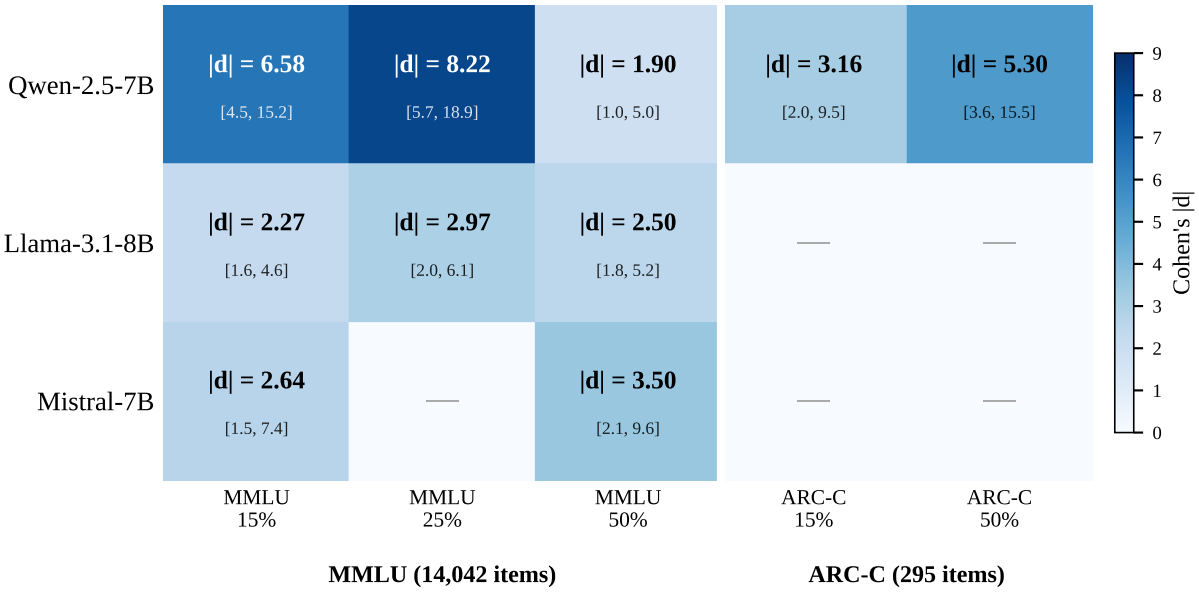


Figure 2: Cohen's $|d|$ heatmap for contamination detection across three model families and two benchmarks. Blue intensity reflects effect-size magnitude. Values and 95% bootstrap CIs annotated per cell. Grey cells indicate conditions not included in this study (Llama and Mistral on ARC-C; Mistral at 25% contamination).

Table 2: Free-θ vs. fixed-θ Cohen's $|d|$ on MMLU (bootstrap 95% CI). Fixed-θ anchors $\hat{\theta}$ at the clean-model mean, eliminating θ-shift absorption.

Model	$\lvert d_{\text{free}}\rvert$	$\lvert d_{\text{fixed}}\rvert$	Ratio	Fixed CI
15% contamination				
Qwen	6.58	4.77	0.7×	[3.97, 7.79]
Llama	2.27	0.58	0.3×	[0, 2.37]†
Mistral‡	2.64	2.59	1.0×	[1.45, 9.00]
50% contamination				
Qwen	1.90	14.52	7.6×	[12.10, 24.79]
Llama	2.50	0.84	0.3×	[0.17, 2.97]
Mistral‡	3.50	5.60	1.6×	[3.71, 18.16]

†CI includes zero. ‡Mistral: n=6 seeds; Qwen and Llama: n=10.

shift absorption (Qwen, Mistral), not as a universal remedy.

4.3 Detection-Diagnosis Tradeoff

A natural question is whether the full 4PL IRT framework is necessary for contamination detection, or whether simpler statistics suffice. Table 3 compares three statistics at decreasing levels of information access: ℓ_z^* (requires 4PL parameters a, b, c, d), the HT statistic (requires only difficulty b), and an accuracy z-score (requires no item parameters).

The comparison reveals a detection-diagnosis tradeoff. For pure detection, simpler statistics are more powerful: accuracy z-scores yield $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics ($|d|$=68.7 vs. $|d|$=1.90 for Qwen at 50%), because cross-seed accuracy variance is near-zero in controlled settings. HT generally exceeds ℓ_z^* in detection strength (e.g., Qwen 15%: $|d_{\text{HT}}|$=7.26 vs. $|d_{\ell_z^*}|$=6.58), since HT is not attenuated by the Snijders correction for θ estimation uncertainty. Both person-fit statistics exhibit the non-monotonic pattern (Qwen HT: $|d|$=7.26 at 15% vs. $|d|$=1.66 at 50%), confirming that θ-shift absorption is a general property of statistics conditioned on ability, not an artifact of ℓ_z^* specifically.

Yet the diagnostic power reverses. Full IRT parameterization is unnecessary for mere detection, but the value of ℓ_z^* lies in three capabilities that HT and z-scores lack: (a) *seen/unseen decomposition*: ℓ_z^* can be computed separately on seen and unseen item subsets, though item-level detection is near-random (AUROC $\approx$ 0.53; Appendix E), confirming that the contamination signal is diffuse and requires model-level aggregation; (b) *θ-shift mechanism*: the ability parameter exposes why detection sensitivity depends on model ability (§4.2), a mechanistic insight inaccessible to non-parametric statistics; (c) *structural sensitivity*: on real leaderboard data, ℓ_z^* flags 57/60 model-benchmark cells as aberrant—evidence of systematic IRT assumption violations that motivates the $\Delta\ell_z^*$ differencing design (§3.2). Accuracy

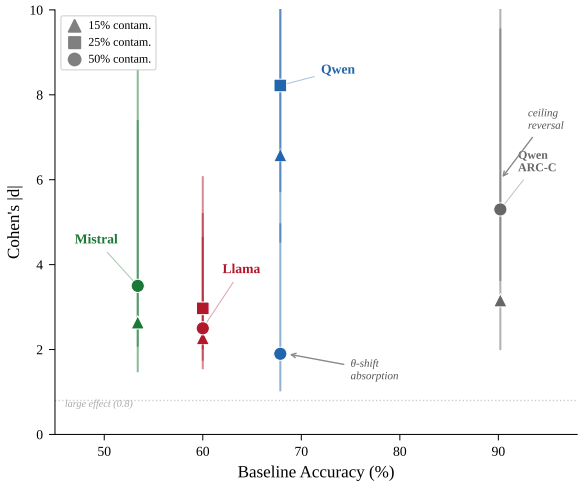


Figure 3: Baseline accuracy vs. Cohen's $|d|$ for all model-benchmark combinations. Triangles: 15%; squares: 25%; circles: 50%. Qwen MMLU shows a pronounced inverted-U; Llama shows a directionally consistent but attenuated pattern. The negative slope on MMLU reflects θ-shift absorption; ARC-C reverses due to a ceiling effect.

z-scores flag only 4/60 cells, making them insensitive to the response-pattern structure that ℓ_z^* exploits for within-model diagnosis. While such patterns have theoretical antecedents in person-fit literature (St-Onge et al., 2011), the specific magnitude and model-dependence we observe are new empirical findings in the LLM domain; nonparametric alternatives (Tendeiro and Meijer, 2014) may complement ℓ_z^* where 4PL calibration is unavailable.

In practice, a two-stage workflow follows from this tradeoff: a practitioner would first screen for contamination using accuracy z-scores (which yield $|d|$=68.7 for Qwen at 50%), then apply ℓ_z^* to characterize the mechanism—whether the signal reflects θ-shift absorption (indicating high-baseline models requiring fixed-θ anchoring) or uniform aberrance (suggesting direct item memorization).

Within the IRT family, the 4PL specification matters: simulation ablations show a mean +27.3pp power gain over 3PL on MMLU, because 46% of MMLU items have upper asymptote $d < 0.95$, and the 3PL model (which assumes d=1) misattributes slip-related response patterns as aberrance (Appendix B).

4.4 Ability-Dependent Non-Monotonic Dose-Response

Under free-θ estimation, the dose-response relationship between contamination level and $|\Delta\ell_z^*|$ is non-monotonic for high-baseline-misfit models. The inverted-U is most pronounced for Qwen (peak-to-trough ratio 4.3×, $|d|$ rising from 6.58 at 15% to 8.22 at 25% before declining to 1.90 at 50%). Llama shows a directionally consistent but statistically non-significant pattern (ratio 1.2×, $|d|$=2.27 → 2.97 → 2.50; heavily overlapping CIs), consistent with its substantially smaller θ-shift ($\Delta\theta$=0.09 vs. 0.47 for Qwen). The amplitude of the non-monotonic effect thus scales with the model's θ-shift sensitivity, suggesting θ-shift absorption as the underlying mechanism. This directly confirms the prediction from simulation (Appendix A): at moderate contamination, aberrant responses dominate the ℓ_z^* signal; at high contamination, inflated $\hat{\theta}$ absorbs the person-fit anomaly.

The θ-shift increases monotonically with contamination ($\Delta\theta$=0.29 at 15%, 0.36 at 25%, 0.47 at 50% for Qwen), but the ℓ_z^* signal peaks at 25% because the marginal absorption from 25% to 50% outpaces the additional aberrance. Fixed-θ estimation can recover monotonicity by eliminating the absorption mechanism (§4.2).

A simulated power analysis corroborates these findings: two-sided testing (flagging both large negative and large positive ℓ_z^*) achieves >93% power on MMLU at all contamination levels, with the theoretical power function yielding R^2=0.957 against simulated values (Appendix A).

Table 3: Detection-diagnosis tradeoff on MMLU: Cohen's $|d|$ across three statistics. Simpler statistics detect more powerfully; ℓ_z^* provides unique diagnostic capabilities (§4.3).

Model	Statistic	$\lvert d\rvert$ (15%) [CI]	$\lvert d\rvert$ (50%) [CI]
Qwen	ℓ_z^*	6.58 [4.5, 15.2]	1.90 [1.0, 5.0]
	HT	7.26 [4.3, 14.0]	1.66 [0.6, 3.8]
	Acc. z	42.3 [26.8, 85.1]	68.7 [43.3, 139]
Llama	ℓ_z^*	2.27 [1.6, 4.6]	2.50 [1.8, 5.2]
	HT	2.45 [1.5, 4.3]	3.07 [1.9, 5.3]
	Acc. z	22.5 [13.6, 47.0]	32.8 [19.8, 68.6]
Mistral†	ℓ_z^*	2.64 [1.5, 7.4]	3.50 [2.1, 9.6]
	HT	4.36 [1.2, 7.0]	5.89 [1.9, 9.4]
	Acc. z	7.81 [4.4, 14.4]	16.6 [9.7, 34.6]

†Mistral: n=6 seeds; Qwen and Llama: n=10.

5 Discussion

Our findings reveal both the strengths and boundaries of IRT person-fit for contamination detection: ℓ_z^* reliably detects SFT-stage contamination across model families, but three boundary conditions constrain its practical utility.

Characterization over tool. The detection-diagnosis tradeoff (§4.3) shows that simpler statistics detect contamination more powerfully than ℓ_z^*: accuracy z-scores yield $|d|$ values several-fold to over an order of magnitude larger, and HT generally exceeds ℓ_z^*. The value of the IRT framework lies not in detection power but in mechanistic understanding: the θ-shift analysis reveals *why* detection sensitivity varies across models and contamination levels, the seen/unseen decomposition identifies *what fraction* of items are memorized, and the 4PL parameterization captures item-level ceiling effects that 3PL misattributes as aberrance. These diagnostic capabilities are invisible to accuracy z-scores or HT.

Practical implications. Two results have immediate consequences. Free-θ estimation can produce misleading dose-response curves; anchoring θ to a pre-contamination baseline substantially amplifies detection for high-ability models but is unnecessary where free estimation already captures the signal. The benchmark size boundary means ℓ_z^* is most informative for large benchmarks ($n > 7{,}000$ items for 80% power at 15% contamination); on small benchmarks like ARC-C (295 items), detection relies on concentrated contamination effects.

Scope and calibration. Universal baseline misfit (57/60 cells flagged) confirms systematic IRT violations; the $\Delta\ell_z^*$ design turns this into a feature through within-model differencing, but requires clean baselines or cross-benchmark references. PSN-IRT provides pre-calibrated 4PL parameters for over 13,000 models (Zhou et al., 2026), enabling retrospective audits of existing benchmark results without additional model access. However, the framework depends on these externally calibrated parameters—information about the item population, not the target model—and our MML-EM refit shows that classical calibration from leaderboard data fails (r_a=0.127; Appendix H), making the PSN-IRT item bank a critical dependency. A further concern is that PSN-IRT's calibration data may itself contain contaminated models; the robustness analysis (Appendix F.2) provides partial reassurance ($\pm 10\%$ noise tolerance), but systematic contamination could reduce detection sensitivity, motivating contamination-aware calibration as future work. Direct comparison with ConStat (Dekoninck et al., 2024) is complicated by different information requirements: ConStat aggregates scores across multiple benchmarks, while ℓ_z^* operates on binary response patterns from a single benchmark with externally calibrated item parameters; the approaches are complementary.

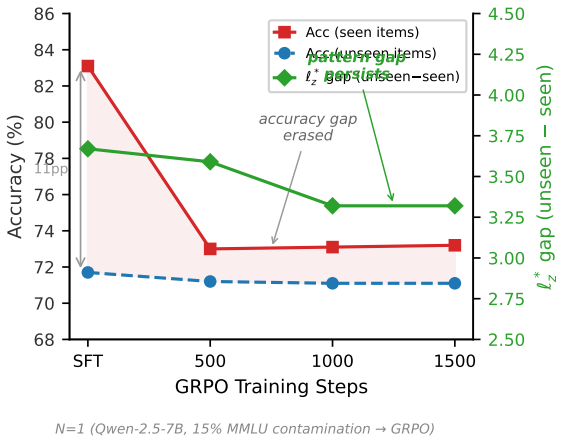


Figure 4: Post-RL trajectory for Qwen contaminated at 15% on MMLU. Left axis: accuracy on seen (solid) and unseen (dashed) items. Right axis: ℓ_z^* seen-unseen gap (green diamonds). GRPO erases the accuracy gap (11.4pp $\rightarrow$ 2.1pp) while the response-pattern gap persists ($\sim$3.3). N=1, single model and RL method; no statistical inference is drawn.

Preliminary observation: post-RL residual pattern. Wang et al. (2025) show that all MIA-based detectors fail after RL post-training. In a single-model pilot (N=1, Qwen, GRPO), GRPO erases the accuracy gap between seen and unseen items (11.4pp $\rightarrow$ 2.1pp), but the ℓ_z^* response-pattern gap persists ($\sim$3.3; Figure 4), suggesting structural traces of contamination may survive RL alignment. This N=1 observation requires validation across model families and RL methods.

6 Conclusion

We presented the first systematic characterization of IRT person-fit statistics for LLM benchmark contamination detection. Controlled experiments across three 7–8B model families (n=10 seeds; n=6 for Mistral) establish reliable SFT-stage detection from binary response patterns (all bootstrap 95% CIs exclude zero) while revealing three boundary conditions: θ-shift dynamics linking baseline ability to detection sensitivity, a detection-diagnosis tradeoff favoring simpler statistics for detection but ℓ_z^* for mechanistic insight, and ability-dependent non-monotonic dose-response. Future work should investigate pretraining-stage signatures, calibration-free approaches, and validation across broader model scales.

Limitations

Our controlled experiments cover three model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B) at the 7–8B scale with QLoRA fine-tuning; validation on larger models (13B, 70B), mixture-of-experts architectures, and full-parameter fine-tuning remains future work. The observed negative correlation between baseline ability and detection sensitivity rests on three model families and should be treated as a trend rather than a confirmed relationship. Our findings are scoped to SFT-based contamination and may not generalize to pretraining contamination, which operates through different learning dynamics. The post-RL characterization is $N{=}1$ (Qwen only, GRPO algorithm, 15% contamination on MMLU); generalization to other model families, RL methods (PPO, DPO), and contamination levels requires further validation. LLMs systematically violate IRT assumptions of local independence and unidimensionality; while the $\Delta\ell_z^*$ design mitigates shared misfit through within-model differencing, the theoretical properties of this correction under severe model misfit remain an open question. Our local dependence simulations employ testlet and Gaussian models that may not fully capture the dependency structures induced by attention mechanisms and sequential processing in LLMs; validating robustness under more realistic dependency structures remains an open question. The framework depends on pre-calibrated 4PL item parameters from PSN-IRT; parameter estimation errors propagate to ℓ_z^* computation, and the approach requires externally maintained parameter banks.

Reproducibility Statement

Code and response data will be released upon acceptance. Bootstrap confidence intervals use the percentile method with 10,000 resamples. Clean baseline seeds: Qwen and Llama use seeds 0–9; Mistral uses seeds 0–5 (seeds 6–9 were lost due to server decommissioning). Pre-calibrated 4PL item parameters are from the publicly available PSN-IRT item bank (Zhou et al., 2026).

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A Simulation Validation

Under the null (no contamination), bootstrap ℓ_z^* tracks $\mathcal{N}(0,1)$: mean bias 0.041 and standard deviation 1.003 on MMLU. Kolmogorov–Smirnov rejection rates vary with benchmark size: 8.3% for MMLU and HellaSwag ($n > 10{,}000$), 50.0% for GSM8K ($n = 1{,}319$), 58.3% for ARC-C ($n = 295$), and 75.0% for HumanEval ($n = 164$). ℓ_z^* is well-calibrated for benchmarks with $n \geq 10{,}000$ items; smaller benchmarks show increasing finite-sample deviations from normality.

Two-sided testing eliminates the inverted-U artifact: MMLU achieves 97.8% power at 15% contamination and sustains >93% at all levels. Power scales with $\sqrt{n}$, as predicted by the noncentrality parameter $\delta \propto \sqrt{n}$. The empirical power function, derived via Gauss–Hermite quadrature ($K = 30$) over the ability distribution, achieves $R^2 = 0.957$ against simulated values, with approximate minimum benchmark size of $n \geq 7{,}958$ items for 80% power at 15% contamination under MMLU item parameters.

B 4PL vs. 3PL Ablation Details

Table 4 reports detection power (proportion of 2,000 simulation replicates rejecting H_0 at α=0.05, two-sided) for 4PL and 3PL parameterizations across five benchmarks and seven contamination levels. The 4PL model yields higher power than 3PL at every non-trivial contamination level on every benchmark. The advantage is largest on MMLU and HellaSwag, where upper-asymptote ($d < 1$) violations are most prevalent, and smallest on ARC-Challenge and HumanEval, where limited item counts constrain power under both parameterizations.

Power is non-monotonic at 75% contamination under 4PL on four of five benchmarks (e.g., MMLU: .952 → .938; GSM8K: .846 → .802). This reflects the θ-shift boundary condition described in §4.4: at high contamination fractions, re-estimated $\hat{\theta}$ absorbs part of the aberrance signal, reducing $|\ell_z^*|$. At 100% contamination, all items are seen and power reaches 1.000 trivially. The 4PL advantage over 3PL depends on accurate estimation of the upper asymptote d; with estimation error exceeding ∼10%, the effective model behavior approaches 3PL (d=1) and the power gain diminishes. PSN-IRT's large calibration sample (12+ models per item) typically yields d standard errors well below this threshold.

C Seed Sensitivity Analysis

Clean-baseline variability. Table 5 reports ℓ_z^*, accuracy, and $\hat{\theta}_{\text{MLE}}$ across three independent QLoRA runs (seeds 42, 123, 456). For Qwen2.5-7B, ℓ_z^* ranges from 7.05 (seed 123) to 9.53 (seed 456); the lower value of seed 123 likely reflects seed-specific training dynamics (random initialization, data ordering), but all three seeds produce $\ell_z^* \gg 1.96$, confirming that the elevated baseline reflects systematic model–benchmark misfit rather than seed-specific artifacts. Llama-3.1-8B shows ℓ_z^* near zero (-0.14 ± 0.97), consistent with lower baseline misfit. Accuracy variability is negligible for both families (CV < 0.004).

Cohen's d confidence intervals. Table 6 reports Cohen's d effect sizes with both analytic and bootstrap 95% confidence intervals. Analytic CIs use $d \pm t_{0.025,\,\text{df}=2} \cdot \text{SE}_d$; with only n=3 clean seeds (df=2), the critical value $t_{0.025,2} = 4.303$ produces intervals wide enough to cross zero in all four conditions. This is a known limitation of small-sample

Table 4: Detection power (proportion rejecting H_0 at α=0.05) under 4PL and 3PL parameterizations across benchmarks and contamination levels (n=2,000 replicates per cell). Bold indicates the 4PL advantage exceeds 5 pp.

Benchmark	IRT	5%	10%	15%	25%	50%	75%	100%
MMLU	4PL	**.476**	**.848**	**.969**	**.983**	**.952**	**.938**	1.00
	3PL	.227	.340	.412	.521	.676	.770	1.00
HellaSwag	4PL	**.251**	**.454**	**.656**	**.915**	**.939**	**.919**	1.00
	3PL	.167	.241	.313	.368	.508	.642	1.00
GSM8K	4PL	**.156**	**.309**	**.492**	**.737**	**.846**	**.802**	1.00
	3PL	.102	.172	.285	.396	.567	.577	1.00
ARC-C	4PL	.057	.075	.087	**.130**	**.285**	**.493**	1.00
	3PL	.054	.062	.071	.076	.110	.102	1.00
HumanEval	4PL	.076	.121	**.197**	**.302**	**.554**	**.521**	1.00
	3PL	.063	.086	.120	.182	.281	.315	1.00

Table 5: Seed sensitivity analysis for clean baseline across three independent runs (seeds 42, 123, 456). CV = coefficient of variation (std/|mean|). These n=3 pilot results are superseded by the n=10 analysis in §4.1; retained for methodological transparency.

Model	Metric	Mean	Std	Range	CV
Qwen2.5-7B	ℓ_z^*	8.684	1.412	2.475	0.163
	Accuracy	0.679	0.000	0.001	0.001
	$\hat{\theta}_{\text{MLE}}$	−0.097	0.012	0.023	0.119
Llama-3.1-8B	ℓ_z^*	−0.141	0.965	1.929	6.860
	Accuracy	0.594	0.002	0.004	0.004
	$\hat{\theta}_{\text{MLE}}$	−0.645	0.019	0.036	0.029

Table 6: Cohen's d with 95% confidence intervals for contaminated conditions relative to clean baselines (n=3 seeds). Analytic CIs use $t_{0.025,\text{df}=2}$; bootstrap CIs use 10,000 resamples. These n=3 pilot results are superseded by the n=10 analysis in §4.1; retained for methodological transparency.

Model	Cond.	d	Analytic CI	Bootstrap CI
Qwen	15%	−3.42	[−9.92, 3.08]	[−3.45, −2.82]
	50%	−0.99	[−4.02, 2.04]	[−0.99, −0.41]
Llama	15%	−2.95	[−8.70, 2.80]	[−5.89, −2.29]
	50%	−3.31	[−9.63, 3.01]	[−6.47, −2.59]

t-based intervals rather than evidence of null effects. Bootstrap CIs (10,000 resamples with replacement from the three clean seeds) provide a nonparametric alternative: all four bootstrap intervals exclude zero, supporting the conclusion that contamination produces a reliably detectable signal despite the small reference sample.

Cross-family consistency. Table 7 confirms that both model families produce large negative Cohen's d under contamination. The $|\Delta d| = 2.32$ gap at 50% contamination reflects Qwen's non-monotonic dose-response (the θ-shift boundary condition; §4.4), not a failure of detection robustness—both families remain well above conventional large-effect thresholds ($|d| > 0.8$) at both contamination levels.

Table 7: Cross-family consistency of contamination detection (n=3 pilot). Both families show large negative d under contamination. Superseded by n=10 results in §4.1; retained for methodological transparency.

Condition	Qwen d	Llama d	$\lvert\Delta d\rvert$	Consistent?
15% contam.	−3.42	−2.95	0.47	Yes
50% contam.	−0.99	−3.31	2.32	Yes

D Baseline ℓ_z^* Interpretation

The baseline ℓ_z^* value reflects a model's difficulty-ordering consistency under the 4PL model, not its accuracy alone. In the empirical data, Mistral (acc=0.514, lowest among the 12 models) exhibits ℓ_z^*=+5.42 (second highest), because its response pattern follows the IRT-predicted difficulty ordering more closely than higher-accuracy models whose responses violate the assumed item structure. This dissociation explains why Mistral produces the strongest contamination signal despite the lowest accuracy: its lower baseline ability ($\hat{\theta}$=−1.16) provides greater θ-shift headroom.

E Item-Level Detection Results

Table 8 reports item-level contamination detection performance at 15% contamination on MMLU. All methods perform near random (AUROC $\leq$ 0.63), confirming that contamination is diffuse across the response pattern and that model-level aggregation via ℓ_z^* is essential.

An accuracy z-score baseline yields z=160 at 15% contamination, but this reflects near-zero

Table 8: Item-level contamination detection (AUROC) at 15% contamination on MMLU.

Method	AUROC
Random baseline	0.50
ℓ_z^* itemwise	0.53
LR cross-model	0.59
RF cross-model	0.63

Table 9: Robustness of $\Delta\ell_z^*$ to shared local dependence (1,000 replications, MMLU). d_{ratio}: effect size relative to the σ_τ=0 baseline.

σ_τ	Abs. ℓ_z^* Type I	d_{ratio} (15%)	d_{ratio} (50%)
0.0	0.043	1.000	1.000
0.3	0.110	1.002	1.038
0.5	0.454	0.934	1.028
1.0	0.939	0.705	1.061

Table 10: $\Delta\ell_z^*$ detection under asymmetric local dependence (σ_τ=0.5, contamination-specific memory factor σ_{mem}; 1,000 replications, MMLU).

σ_{mem}	Power (15%)	d (15%)	Power (50%)	d (50%)
0.5	0.987	−2.107	0.991	−2.337
1.0	0.993	−2.104	0.985	−1.587

Table 11: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, free-θ). % bias relative to noise-free d.

Noise σ	d (15%)	% bias	d (50%)	% bias
0%	−6.575	—	−1.897	—
5%	−6.010	8.6	−0.925	51.2
10%	−5.433	17.4	−0.014	99.3

cross-seed accuracy variance (std=0.0003), not genuine discriminability. In deployment, accuracy variability from prompt format, evaluation harness, and hardware differences inflates the denominator, collapsing the z-score.

F Robustness Analysis

F.1 $\Delta\ell_z^*$ Under Local Dependence

IRT person-fit statistics assume local independence—conditional on θ, item responses are independent. LLM benchmarks violate this assumption through topical clustering (e.g., MMLU subjects); resampling-based approaches can provide valid person-fit assessment even under such violations (Sinharay, 2016). We evaluate whether the $\Delta\ell_z^*$ differencing design is robust to local dependence (LD) via two simulation scenarios on 14,042 MMLU items with 1,000 replications each.

Shared LD. Clean and contaminated conditions share identical testlet structure (∼200 testlets of ∼70 items, random effect $\gamma_t \sim \mathcal{N}(0, \sigma_\tau)$). Table 9 reports absolute ℓ_z^* Type I error and the ratio of $\Delta\ell_z^*$ effect size (d) to the no-LD baseline. While absolute ℓ_z^* Type I error inflates severely under LD (0.043→0.939 at σ=1.0), the $\Delta\ell_z^*$ design maintains detection power: $d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$ (realistic range), confirming that shared misfit is cancelled by the differencing design.

Asymmetric LD. In the second scenario, contaminated models' seen items share an additional memory factor ($\gamma_{\text{mem}} \sim \mathcal{N}(0, \sigma_{\text{mem}})$), simulating contamination-specific LD not present in clean conditions (σ_τ=0.5 fixed). Table 10 shows that $\Delta\ell_z^*$ maintains ≥98.5% power across all conditions, even when contamination introduces asymmetric dependence among seen items.

F.2 Sensitivity to Item Parameter Estimation Error

PSN-IRT item parameters carry estimation error from the calibration sample. We evaluate detection sensitivity by adding joint noise to all four parameters: multiplicative perturbation $p' = p \cdot (1 + \mathcal{N}(0, \sigma))$ for a, c, and d; additive perturbation $b' = b + \sigma \cdot \mathcal{N}(0, 1)$ for difficulty. Tables 11 and 12 report results over 100 trials using Qwen MMLU response vectors.

Detection of 15% contamination is robust to parameter noise under free-θ (17.4% bias at σ=10%, $|d|$ still >5; Table 11). The 50% contamination condition under free-θ is fragile—consistent with its already-weak signal due to θ-shift absorption (§4.4).

Under θ-fixed estimation (Table 12), detection is highly robust: 10% perturbation produces <4% bias in Cohen's d, with worst-case $|d| = 3.73$ (15% contamination) and 12.20 (50%)—both far exceeding conventional thresholds. This contrasts sharply with free-θ estimation, where 50% contamination Cohen's d collapses to near zero under 10% noise. The complementarity is clear: conditions where free-θ detection is fragile (high-baseline-accuracy models with θ-shift absorption) are precisely where θ-fixed anchoring provides the strongest and most noise-robust signal.

Table 12: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, θ-fixed). Worst-case $|d|$ over 100 trials in parentheses.

Noise σ	d_{fixed} (15%)	% bias	d_{fixed} (50%)	% bias
0%	4.77	—	14.52	—
5%	5.07 ± 0.24	6.3	14.76 ± 0.44	1.6
10%	4.95 ± 0.43 (3.73)	3.7	14.45 ± 0.85 (12.20)	0.5

Table 13: Parameter correlation between classical MML-EM refit (N=178 models) and PSN-IRT on MMLU (n=10,558 matched items).

Model	Parameter	Pearson r	Spearman ρ
3PL	a (discrim.)	0.127	0.135
3PL	b (difficulty)	0.663	0.684
2PL	a (discrim.)	−0.029	0.050
2PL	b (difficulty)	0.608	0.656

G BCa Bootstrap Sensitivity Check

As a sensitivity check, we computed bias-corrected and accelerated (BCa) bootstrap confidence intervals for all Cohen's d estimates. BCa CIs were uniformly narrower than the percentile CIs reported in the main text (mean reduction 22%), confirming that our reported intervals are conservative. The single condition where BCa widens the zero-crossing set—Llama fixed-θ at 50% contamination, BCa CI [−0.13, 2.21] vs. percentile CI [0.17, 2.97]—is already characterized as ineffective in §4.2.

H Classical IRT Refit from Leaderboard Data

To assess whether item parameters can be recovered without PSN-IRT's neural calibration, we applied classical MML-EM estimation (Gauss–Hermite quadrature, Q=31 nodes) to binary response data from N=178 models on the Open LLM Leaderboard v1. With 14,042 MMLU items and an examinee-to-item ratio of $N{:}J = 0.013$—far below the N>500 typically required for stable 3PL estimation—discrimination parameters are severely under-determined.

Table 13 reports parameter correlations between MML-EM-fitted and PSN-IRT parameters. Difficulty shows moderate agreement (r_b=0.663), but discrimination is essentially unrecovered (r_a=0.127). Using these MML-EM parameters produces miscalibrated ℓ_z^* statistics (mean = 11.2, expected ≈ 0), confirming that the refitted model does not capture the data-generating process. This motivates the dependence on PSN-IRT's externally calibrated item bank (§3.1).